SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA1522

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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I Lakshmi(192373066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _______________

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage

Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Definition: Cloud computing is the delivery of computing services such as servers, storage, databases, networking, and software over the internet on a pay-as-you-use basis.

Characteristics:

- On-Demand Self-Service –Users can access computing resources whenever needed without human intervention.
  Example: Launching a new server instance in minutes through a web portal.
- Broad Network Access –Services are available over the internet and can be accessed from laptops, mobiles, and tablets.
  -This contrasts with traditional on-premise setups, which often require local access
- Resource Pooling –Resources are shared among multiple users using a multi-tenant model.
  -Resources are dynamically assigned and reassigned based on demand, ensuring efficiency and scalability.
- Rapid Elasticity – Resources can be quickly increased or decreased based on demand.
  - For example, an e-commerce site can automatically add servers during peak shopping seasons and scale back afterward.

2.Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Hardware Evolution:

- Mainframe Computers
- Minicomputers
- Personal Computers
- Client-Server Systems
- Virtualized Data Centers

Internet & Software Evolution

- Internet
- Web Applications
- Web Services
- Virtualization
- Cloud Computing

How It Enabled Cloud Platforms:

- Faster internet enabled remote access.
- Virtualization allowed efficient resource sharing.
- Web technologies made applications available online.
- Together, these led to modern cloud platforms like AWS, Azure, and Google Cloud.

3.Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Server Virtualization: It divides one physical server into multiple virtual machines (VMs), each running its own operating system.

Virtualization vs Cloud Computing

Virtualization

- Enabling technology
- Creates virtual machines
- Focuses on hardware utilization

Cloud Computing

- Service delivery model
- Delivers services over the internet
- Focuses on providing on-demand services

Role: Virtualization improves resource utilization, scalability, and cost efficiency, making cloud computing possible.

4.Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications

SOAP(Simple Object Access Protocol): A protocol for exchanging structured information in web services using XML.

- Protocol
- Uses XML only
- More secure and reliable
- Complex communication Simple
- Suitable for enterprise applications

REST(Representational State Transfer): An architectural style for designing networked applications, using standard HTTP methods**.**

- Architectural style
- Uses JSON, XML, HTML, etc.
- Faster and lightweight
- HTTP communication
- Best for cloud and web applications

5.Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

- IaaS (Infrastructure as a Service):Provides virtual servers, storage, and networking. Eg.Amazon EC2

  Use Case: Startups hosting websites without investing in hardware.

- PaaS (Platform as a Service) :Provides a platform to develop and deploy applications.Eg.Google App Engine

  Use Case: Building web apps quickly without worrying about server setup.

- SaaS (Software as a Service):Provides ready-to-use software over the internet.

  Eg. Gmail

  Use Case: Companies managing sales pipelines without installing software locally.

- XaaS (Anything as a Service):Offers any IT service through the cloud.

  Eg.Microsoft 365

Use Case: Individuals storing files in the cloud or developers integrating AI features without building models themselves.

THANKYOU